

KOKSHETAU, Kazakhstan

WMO: 288790

Lat: 53.3169N

Lon: 69.3772E

Elev: 229

StdP: 98.60

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.4	-28.8	-35.6	0.1	-32.4	-32.0	0.2	-28.7	15.5	-8.1	14.3	-6.2	1.6	250	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	31.1	18.4	29.3	18.2	27.4	17.6	20.4	27.3	19.6	26.4	18.8	25.2	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.4	22.9	17.2	12.7	22.2	16.3	11.9	21.7	59.8	27.4	56.9	26.2	54.2	25.1	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.6	11.7	9.4	DB	-34.9	33.6	3.9	2.0	-37.7	35.0	-40.0	36.2	-42.1	37.3	-45.0	38.7	(4)
(5)			WB	-35.1	22.2	4.2	1.3	-38.1	23.1	-40.6	23.9	-43.0	24.6	-46.0	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.4	-15.1	-13.7	-5.4	5.3	13.1	18.5	19.4	18.0	12.2	4.8	-5.0	-11.9	(6)	
	DBStd	13.80	7.17	7.10	6.09	5.81	5.00	4.28	3.63	4.24	4.85	4.98	7.58	7.09	(7)	
	HDD10.0	3441	777	662	478	159	28	2	0	1	30	174	452	678	(8)	
	HDD18.3	5619	1036	896	736	390	173	49	31	59	192	420	702	936	(9)	
	CDD10.0	1049	0	0	0	20	122	257	292	249	96	12	0	0	(10)	
	CDD18.3	185	0	0	0	0	9	54	64	49	8	0	0	0	(11)	
	CDH23.3	1812	0	0	0	5	131	539	595	456	86	0	0	0	(12)	
	CDH26.7	542	0	0	0	1	28	172	178	139	24	0	0	0	(13)	
(14) Wind	WSAvg	4.0	3.6	4.1	4.5	4.3	4.4	3.7	3.3	3.3	3.7	4.2	4.5	4.4	(14)	
(15) Precipitation	PrecAvg	310	11	11	10	18	29	39	73	45	24	21	16	11	(15)	
	PrecMax	524	49	40	27	60	61	104	196	123	66	73	60	33	(16)	
	PrecMin	201	1	0	0	0	1	3	21	2	7	0	0	0	(17)	
	PrecStd	88	9	8	7	13	17	24	47	29	14	15	11	7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	1.6	9.9	23.3	29.6	32.8	33.4	32.3	29.7	20.8	9.0	1.9	(19)	
		MCWB	0.5	0.2	5.1	12.6	16.1	18.1	19.4	19.1	16.8	11.6	5.9	0.4	(20)	
	2%	DB	-0.8	-0.2	5.6	19.9	26.6	30.7	30.7	30.1	25.7	17.5	6.7	0.5	(21)	
		MCWB	-1.8	-1.4	2.7	11.1	15.1	17.9	19.0	18.7	14.7	10.2	4.4	-0.5	(22)	
	5%	DB	-2.7	-1.7	3.7	16.9	24.3	28.7	28.5	28.1	22.7	14.7	4.9	-0.8	(23)	
		MCWB	-3.6	-2.9	1.5	9.5	14.2	17.9	18.7	18.2	13.9	8.7	3.1	-1.9	(24)	
	10%	DB	-4.6	-3.6	2.2	14.1	22.2	26.5	26.6	25.9	20.0	12.1	3.3	-2.5	(25)	
		MCWB	-5.4	-4.7	0.7	8.2	13.5	17.0	18.4	17.3	12.7	7.7	1.6	-3.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	0.5	5.3	13.4	18.2	21.0	21.6	21.2	17.8	12.4	6.9	0.7	(27)	
		MCDB	1.4	1.6	8.9	21.2	26.3	28.6	27.9	28.6	27.0	18.6	8.7	1.5	(28)	
	2%	WB	-1.8	-1.2	3.0	11.6	16.4	19.6	20.7	20.0	15.8	10.9	4.6	-0.4	(29)	
		MCDB	-0.8	-0.2	5.3	18.2	24.0	27.3	27.1	27.1	23.2	16.2	6.4	0.5	(30)	
	5%	WB	-3.6	-2.7	1.8	10.2	15.1	18.7	19.9	19.0	14.6	9.3	3.2	-1.9	(31)	
		MCDB	-2.8	-1.7	3.4	16.0	22.9	26.1	26.4	25.8	21.2	13.8	4.8	-0.8	(32)	
	10%	WB	-5.4	-4.7	0.7	8.6	14.0	17.8	19.1	18.0	13.3	7.9	1.7	-3.6	(33)	
		MCDB	-4.6	-3.7	2.3	13.9	20.8	24.6	24.9	24.5	19.1	11.7	3.1	-2.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.4	8.4	9.6	11.4	10.8	10.3	10.6	9.7	7.5	6.2	7.0	(35)	
		MCDBR	6.5	6.7	8.0	13.9	14.5	13.6	13.4	13.8	13.8	12.4	6.6	7.0	(36)	
	5% WB	MCWBR	6.4	6.0	5.9	7.2	6.8	5.6	5.4	5.9	6.4	7.0	5.2	6.5	(37)	
		MCWBR	6.6	6.4	7.3	12.9	13.1	11.8	11.3	12.2	11.9	11.3	6.2	6.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.283	0.312	0.369	0.406	0.404	0.407	0.429	0.421	0.367	0.346	0.318	0.283	(40)	
		taud	2.076	2.042	1.989	2.139	2.222	2.293	2.221	2.227	2.332	2.282	2.148	2.065	(41)	
	Ebn at Noon		658	760	789	819	844	845	815	797	801	727	608	570	(42)	
		Edh at Noon	72	104	137	135	131	124	131	123	98	82	68	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.17	2.29	3.75	4.63	5.65	6.03	5.55	4.75	3.41	1.80	0.90	0.78	(44)	
	RadStd		0.08	0.09	0.41	0.48	0.47	0.48	0.43	0.39	0.25	0.27	0.13	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page